

UniMate AI

AI-Powered Smart University Platform

Project Documentation

Document No.: 04

Software Requirements Specification (SRS)

Part VI

Requirements Traceability Matrix (RTM)

IEEE 29148 Inspired

Version: 1.0

Prepared By

Abdullah Shabat

Graduation Project

Faculty of Information Technology

Department of Distributed Computing & Smart Device Applications

Islamic University of Gaza

Academic Year
2025–2026

1. Introduction

The Requirements Traceability Matrix (RTM) is a document used to ensure that every system requirement is properly tracked throughout the software development lifecycle.

It creates a relationship between:

- User Requirements.
- Functional Requirements.
- System Modules.
- Implementation Components.
- Testing Procedures.

The purpose of this matrix is to guarantee that no requirement is ignored during development and testing.

2. Traceability Objectives

The RTM aims to:

- Ensure complete requirement coverage.
- Support software testing activities.
- Track implementation progress.
- Identify missing features.
- Simplify future maintenance.

3. Traceability Matrix Structure

Each requirement is mapped using the following structure:

Field	Description
Requirement ID	Unique requirement identifier
Requirement Description	Requirement summary
Module	Related system module
User Role	Responsible actor

Field	Description
Implementation Component	Software component
Test Case	Validation method

4. Requirements Traceability Matrix

Authentication Module					
ID	Requirement	Module	Role	Implementation	Test Case
FR-001	User Registration	Authentication	Student / Doctor	Firebase Authentication + Backend API	TC-001 Register New User
FR-002	User Login	Authentication	All Users	Firebase Auth + JWT	TC-002 Login Validation
FR-003	User Logout	Authentication	All Users	Token Management	TC-003 Logout Test
FR-004	Password Reset	Authentication	All Users	Firebase Password Recovery	TC-004 Reset Password
FR-005	Role Authorization	Security	All Users	RBAC Middleware	TC-005 Role Access Test
User Profile Module					
ID	Requirement	Module	Role	Implementation	Test Case
FR-006	View Profile	User Management	All Users	User API + Database	TC-006 View Profile
FR-007	Update Profile	User Management	All Users	Profile API	TC-007 Update Profile
Course Management Module					
ID	Requirement	Module	Role	Implementation	Test Case
FR-008	Create Course	Course Management	Doctor	Course API + PostgreSQL	TC-008 Create Course
FR-009	View Courses	Course Management	Student / Doctor	Course API	TC-009 Retrieve Courses
FR-010	Update Course	Course Management	Doctor	Course Service	TC-010 Update Course
FR-011	Delete Course	Course Management	Doctor/Admin	Course Service	TC-011 Delete Course

Learning Materials Module

ID	Requirement	Module	Role	Implementation	Test Case
FR-012	Upload Materials	File Management	Doctor	Supabase Storage	TC-012 Upload File
FR-013	Download Materials	File Management	Student	Storage API	TC-013 Download File
FR-014	Manage Files	File Management	Doctor	File Service	TC-014 File Management

Assignment Module

ID	Requirement	Module	Role	Implementation	Test Case
FR-015	Create Assignment	Assignment Module	Doctor	Assignment API	TC-015 Create Assignment
FR-016	Submit Assignment	Assignment Module	Student	Submission API	TC-016 Submit Assignment
FR-017	View Status	Assignment Module	Student	Assignment Service	TC-017 Check Status
FR-018	Grade Assignment	Assignment Module	Doctor	Grading Service	TC-018 Grade Submission

AI Module

ID	Requirement	Module	Role	Implementation	Test Case
FR-019	AI Chat	AI Module	Student / Doctor	Gemini API Integration	TC-019 AI Conversation
FR-020	Summarization	AI Module	Student / Doctor	Gemini Processing Service	TC-020 Generate Summary
FR-021	Chat History	AI Module	Student / Doctor	AI Chat Database	TC-021 Retrieve History

File Management Module

ID	Requirement	Module	Role	Implementation	Test Case
FR-022	Upload File	Storage Module	Authorized Users	Supabase Storage	TC-022 File Upload
FR-023	Secure File Access	Security Module	Authorized Users	Access Control	TC-023 Permission Check
FR-024	Delete File	Storage Module	File Owner	Storage Service	TC-024 Delete File

Dashboard Module

ID	Requirement	Module	Role	Implementation	Test Case
FR-025	Admin Dashboard	Dashboard	Admin	React Dashboard	TC-025 Admin View
FR-026	User Management	Dashboard	Admin	User APIs	TC-026 Manage Users
FR-027	Doctor Dashboard	Dashboard	Doctor	React Dashboard	TC-027 Doctor View

Notification Module

ID	Requirement	Module	Role	Implementation	Test Case
FR-028	Send Notifications	Notification	System	Firebase Cloud Messaging	TC-028 Send Notification
FR-029	View Notifications	Notification	Users	Notification API	TC-029 View Notifications

5. Non-Functional Requirements Traceability

ID	Requirement	Category	Implementation	Validation
NFR-001	API Performance	Performance	ASP.NET Optimization	Load Testing
NFR-005	Secure Authentication	Security	Firebase + JWT	Security Testing
NFR-006	Authorization Control	Security	RBAC Middleware	Access Testing
NFR-013	User Scalability	Scalability	Clean Architecture	Stress Testing
NFR-016	User Friendly Interface	Usability	Material Design	User Evaluation
NFR-019	Clean Architecture	Maintainability	Layered Architecture	Code Review
NFR-021	Version Control	Maintainability	Git Workflow	Repository Review

6. Requirement Coverage Summary

Category	Total Requirements	Covered
Functional Requirements	29	29
Non-Functional Requirements	28	28
External Interfaces	All Defined	Complete
Constraints	Defined	Complete

7. Implementation Tracking

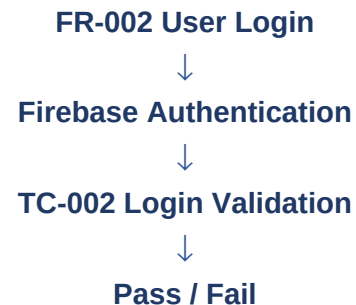
Development progress will be tracked using the following states:

Status	Meaning
Planned	Requirement identified
In Progress	Currently under development
Completed	Implemented successfully
Tested	Verified through testing
Approved	Accepted by supervisor

8. Testing Relationship

Each requirement must have at least one validation method.

Example:



9. Future Expansion Traceability

Future requirements will follow the same structure:

Future Requirement	Module	Version
AI Quiz Generator	AI Module	Version 2.0
Smart Study Planner	Planning Module	Version 2.0

Future Requirement	Module	Version
Analytics Dashboard	Dashboard	Version 3.0

10. Conclusion

The Requirements Traceability Matrix completes the Software Requirements Specification document by connecting system requirements with implementation and testing activities.

Through this approach, UniMate AI maintains clear control over project development, reduces missing requirements, and ensures that every implemented feature contributes directly to the project objectives.